



Exploring the impact of generative artificial intelligence on students' learning outcomes: a meta-analysis

Yinkun Zhu¹ · Qiwen Liu¹ · Li Zhao¹ 

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Abstract

Generative artificial intelligence (GAI) has brought new ideas for optimizing students' learning. Despite increasing attention on the effects of GAI on learning outcomes (LO), research results are inconsistent. While GAI's educational benefits are qualitatively described, there is substantial debate about its actual impact on students' LO. The study sought to quantify GAI's impact on students' LO, evaluating its overall and average effects, and examining four key moderating factors: functional types of GAI, educational levels, intervention duration, and knowledge domains. Based on the screening criteria, 26 empirical studies were selected from 5,887 peer-reviewed papers. Two researchers collaboratively completed the literature screening and coding process. The research employed a meta-analytic method to calculate the impact of GAI on learners' LO, and examined four moderating factors. GAI exerts a significant but small overall effect on students' LO ($g=0.392$), with varying impacts on physical ($g=0.701$), social-emotional ($g=0.347$), and intellectual ($g=0.372$) outcomes. The changes of GAI's functional types have no significant effect on LO, but three other moderating factors do, showing significant statistical differences. GAI more significantly impacts primary school students, especially in supporting their intellectual and social-emotional outcomes. Longer interventions have a greater effect on LO than short ones, particularly intellectual and physical outcomes. GAI's effects vary across knowledge domains, possibly due to its adaptability in different subjects. Long-term GAI in higher education boosts intellectual and physical outcomes, especially in education and humanities and arts, while short-term use in primary education enhances social-emotional outcomes. Integrating diverse learning components and adjusting GAI implementation parameters can optimize its effectiveness in terms of enhancing LO across different levels of education.

Keywords Generative artificial intelligence · ChatGPT · Chatbots · Learning outcomes · Meta-analysis

Extended author information available on the last page of the article

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1 Introduction

1.1 The effects of GAI on students' LO

As information technology advances, generative artificial intelligence (GAI) has been progressively entering the purview of the general public. GAI includes but is not limited to models such as ChatGPT, DALL-E, Bard, and chatbots, which show great promise due to their multi-party applications (Sai et al., 2024). The strong capabilities of text generation, translation, and dialogue make it possible for GAI to assist in assessment practices (Nikolic et al., 2023), gradually making it a widely used new tool in education (Crompton & Burke, 2023).

In the educational context, GAI models such as artificial intelligence chatbots can serve as teaching assistants or instructors, providing guidance throughout the students' learning process to ensure a more positive experience (Wollny et al., 2021). These models are mostly designed in accordance with systematic theories, adopting the architecture of adaptive learning systems, and taking into account the complexity and entirety of learning. For example, dialogue-based intelligent tutoring systems like AutoTutor can track student progress, adapt conversations, and react to their emotions (Graesser, 2016). GAI tutors, powered by large language models, accurately understand and respond to student inquiries (Lee, 2024). Compared to non-AI technologies, GAI offers an immediate and personalized learning environment that can more effectively motivate students to engage in self-regulated learning (Chiu, 2024). By analyzing individual learner data to generate customized educational content (Shahzad et al., 2024), GAI can offer personalized instruction, assisting students in managing their educational endeavors, establishing objectives, and evaluating the advancement towards those objectives (Kong & Yang, 2024). This aligns with self-regulation theory, which posits that learners who actively monitor and control their cognition, motivation, and behavior will benefit in terms of their learning (Zimmerman, 2002). GAI tailors its output to individual student dynamics, empowering self-directed learning and clear expression of their needs during interactions.

As the functionalities of GAI continue to be developed, it is increasingly seen as an evaluative tool capable of providing feedback on educational teaching and student learning. In particular, when leveraging and synthesizing diverse tracking data, GAI enhances the precision of knowledge, skills, and immediate evaluations necessary for self-directed learning, empowering students to fine-tune their strategies based on a critical analysis of their performance (Giannakos et al., 2024). This dynamic assessment of students' learning processes and outcomes, such as writing skills (Boudouaia et al., 2024) and reading skills (Behforouz & Ghaithi, 2024), demonstrates the vast potential of GAI in transforming teaching methodologies and elevating learning outcomes (LO).

Currently, a growing body of research is exploring the impact of GAI on student learning (Jeon, 2022). Evidence suggests that it can enhance students' intellectual LO, such as academic writing performance (Kim et al., 2024) and creative thinking skills (Essel et al., 2024). By creating learning environments tailored to students' needs, GAI enhances the educational experience (Kurtz et al., 2024). In these settings, GAI's content and dialogue generation capabilities can boost student motiva-

tion (Chen & Hou, 2024) and situational interest (Liu et al., 2024), offering emotional support.

GAI's educational role is underpinned by Cognitive Load Theory (Sweller, 2011), which promotes learning designs that reduce cognitive strain on working memory. In the context of GAI, this involves systems presenting complex information in structured, multimodal formats to decrease extraneous cognitive load, facilitating learners to focus on pivotal educational objectives and enhance understanding and retention, which are crucial for impactful learning. For instance, in science, GAI's answers and feedback that are closely aligned with students' cognitive levels significantly reduce their cognitive load and promote more effective learning behaviors (Chen & Chang, 2024). Similarly, in language learning, GAI's integration with poetry and language through multimedia resources, such as Tang poetry videos, decreases cognitive load and improves comprehension (Chen & Wu, 2024).

Nonetheless, numerous studies have also found that the utilization of GAI does not significantly enhance students' motivation (Kumar, 2021), confidence (Han et al., 2022), or learning engagement (Liu et al., 2022). Even when it comes to practical skills, the utility of GAI for students is questionable. For instance, Fathi et al. (2024) pointed out that GAI provides innovative opportunities to improve students' oral English skills, fully training them in fluency, coherence and accuracy. However, another study found that students who used GAI did not perform as well as those who did not use it in developing their programming skills (Kosar et al., 2024). It is evident that the effects of GAI on students' LO may be more complex than we think. Its effects on various categories of LO may also vary.

To this end, this study introduced Salas-Pilco's (2020) classification framework, which categorizes LO into physical outcomes (PO), social-emotional outcomes (SO), and intellectual outcomes (IO). PO is related to motor skills, which require physical participation and coordination. For instance, the previously mentioned oral skills fall under PO. SO is associated with learners' self-efficacy, motivation, attitudes, and values, mainly connecting with the learners' affective domain. In the cognitive domain, LO is primarily IO, which relates to learners' psychological cognitive results, such as their academic performance, memory levels, and so on. These three types of LO cover all kinds of academic achievements, academic performance, and learning experiences in education, helping researchers make a finer division of LO to discuss the specific manifestations of GAI in students' learning.

1.2 Factors moderating the impact of GAI on students' LO

In the extensive reading of GAI-themed research, we discovered that the effect of GAI on students' LO is not uniform (e.g., Essel et al., 2022; Yin et al., 2021), which may be related to some parameter differences in the specific implementation process, especially in terms of functional types of GAI, educational levels, intervention duration, and knowledge domains.

In the preceding discussion, the widespread application of GAI's personalized support, assessment, and intelligent tutoring functions in education has been acknowledged. However, in the actual intervention, the variability in GAI functionalities may influence students' LO. Brod (2024) pointed out that GAI can support

personalized learning combined with teacher tutoring to enhance learners' learning output, while another study found that, as an intelligent tutoring system, GAI may have some obstacles in assisting students in the learning process, failing to truly improve students' problem-solving abilities, which leads to less than ideal LO (Liao et al., 2024). Previous research has found a phenomenon whereby GAI performs more effectively in higher education (Chaudhry et al., 2023), while its effectiveness in primary and secondary education is not significant (Jauhiainen & Guerra, 2023). One study pointed out that in high school geometry problems, GAI has the capability to provide correct solutions and may offer students wiser approaches to problem solving (Parra et al., 2024). This implies that the effectiveness of GAI also varies among student populations at different educational levels.

The duration of tool usage has always been considered a critical factor affecting the effectiveness of interventions. Some researchers have suggested exploring the enduring effects of GAI on students' LO, in addition to investigating the short-term changes in student learning under GAI intervention (Bahroun et al., 2023). For example, no significant differences in educational achievement were found after a 40-minute GAI session when comparing the intervention cohort to the control cohort (Yin et al., 2021). However, after 16 weeks of interaction with a chatbot, students in the GAI interaction team demonstrated a marked improvement in performance than others in the teacher interaction team (Essel et al., 2022). We also observed that the utilization of GAI in science does not effectively enhance students' programming skills (Kosar et al., 2024). This may be related to knowledge domains in which GAI is used, as it has been found to significantly enhance students' language abilities in humanities and arts (Fathi et al., 2024). Additionally, in terms of social skills, interacting with conversational GAI has been shown to significantly improve students' communicative abilities and willingness to engage (C. Wang et al., 2024). The inconsistent findings among these studies prompted us to further consider the moderating effects of functional types of GAI, educational levels, intervention duration, and knowledge domains on the impact of GAI on students' LO.

1.3 Previous meta-analyses of GAI in student learning

It is considered beneficial to understand how to incorporate GAI into the assessment, teaching, and learning processes within the curriculum (Ogunleye et al., 2024). However, the divergence in findings from empirical evidence on the influence of GAI on students' LO calls for a more objective and holistic approach to evaluating the use of GAI. Meta-analysis serves as a robust method for mitigating the potential biases present in traditional narrative syntheses (Pickering & Byrne, 2014). Based on the raw data collected from a broad spectrum of experimental analyses, investigators can measure the magnitude of the intervention's efficacy (Bai et al., 2020). Employing this method, more robust and persuasive conclusions about the degree of intervention effectiveness can be drawn (Garrido et al., 2018), providing a way to reconcile the conflicting results regarding the impact of GAI. Wu and Yu's (2023) meta-analysis found that artificial intelligence chatbots in GAI exert a beneficial impact on students' grades, motivation, and interest, with differences observed in educational levels and intervention times. Wang et al. (2023) observed that educational robots exert a mod-

erate yet significant positive influence on students' LO. In their analysis, educational level and assessment type were significantly correlated with LO, while moderating factors such as subject area, intervention duration, and types of robots were not significant. Similarly, Alemdag (2023) conducted a meta-analysis and found that the impact of chatbots on student learning is significant and moderate. However, their study only focused on one category of GAI, while other studies have investigated a specific context only. For instance, F. Wang et al. (2024) concentrated on language acquisition and, via a meta-analysis encompassing 28 studies, established the positive, albeit minor, effect of ChatGPT on language learning performance. Zhang et al. (2023) validated the moderate effect of chatbots on language learning and explored nine moderating variables such as educational level and teaching duration. Sun and Zhou (2024), in their meta-analysis, demonstrated that GAI significantly improved students' academic performance with a medium effect size, but their study was limited to the college student population. Another review also suggested that while the integration of GAI may improve students' language learning performance, it may also correspondingly increase the extraneous cognitive load on students (Huang et al., 2022).

Despite the existing research providing evidence of the effects of chatbots, ChatGPT, and the like on students' learning performance, the impact of GAI on students' LO has not been thoroughly examined. Limited studies have investigated the influence of GAI on more nuanced categories of LO, considering dimensions such as physical, social-emotional, and intellectual aspects. Previous reviews have undertaken extensive qualitative assessments and structured content analyses regarding the role of GAI in educational contexts. A meta-analysis would adeptly bridge the gap in researchers' comprehension of GAI's impact on students' LO, offering empirical data to substantiate their assertions and discussions. Therefore, this study endeavored to integrate various subcategories of GAI, including artificial intelligence chatbots and ChatGPT, and leveraging the framework proposed by Salas-Pilco (2020), it investigated LO associated with each category, aiming to offer a more holistic view of GAI's influence on LO. A meta-analysis was undertaken in this study to quantitatively assess the impact of GAI on students' LO, thereby offering insights to inform future GAI design and its integration to foster student growth.

1.4 Current study

The ambiguity of the impact of GAI on students' LO prompted us to conduct a meta-analysis to amalgamate data across various investigations, revealing the overall and average effects of GAI on various aspects of students' LO. By analyzing moderating variables such as functional types of GAI, educational levels, intervention duration, and knowledge domains, the study aids researchers in understanding how these factors influence students' LO, thereby providing guidance for educational practice. In particular, this study was designed to tackle the ensuing questions.

RQ1. What is the overall impact of applying GAI on students' LO, including their PO, SO and IO?

RQ2. How do factors such as the functional types of GAI, educational levels, intervention duration, and knowledge domains moderate the impact of GAI on students' LO?

2 Method

2.1 Search strategy

Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement as established by Page et al. (2021), this study was conducted to ensure that the research findings are scientifically rigorous and reliable. To encompass an extensive array of literature so as to uphold the scientific credibility of the research results, an exhaustive and meticulous literature search was undertaken across four prestigious databases, namely the Web of Science, Science Direct, Scopus, and IEEE Xplore. Given that GPT-3 was first introduced in 2020 and has attracted widespread attention in education (Dwivedi et al., 2023), the literature selection time was set from 2020 to 2024. By examining the classification of GAI, the following search term was used: ((GPT) OR (agent) OR (OpenAI) OR ("large language models") OR ("Generative AI") OR ("generative artificial intelligence") OR (Chatbot) OR (GAI) OR (GenAI)) AND (("learning outcome") OR ("learning performance") OR ("learning effect") OR ("academic performance") OR ("learning achievement")). A systematic search was conducted in August, 2024. When searching, we made use of some restrictions (i.e., "English", "Article") in the databases to focus on the available literature in order to facilitate our further screening. Given the global predominance of English and challenges associated with linguistic dissemination and resource accessibility, this research focused exclusively on journals published in English to facilitate comprehensive literature management and analysis. Based on the above search parameters in titles, abstracts, and keywords, a total of 5,887 articles were yielded, with 3,610 from the Web of Science, 253 from Science Direct, 1,700 from Scopus, and 324 from IEEE Xplore. The search process is illustrated in Fig. 1.

2.2 Study selection

After removing duplicate literature, 3,313 results remained. To guarantee that the selected studies were pertinent to our research theme, the inclusion of studies was predicated on the fulfillment of the following criteria: (1) the study's primary focus must be on GAI's influence on students' LO; (2) the study must be a randomized controlled trial, including an intervention group that used GAI and a comparison group that did not; (3) the complete study document must be accessible to extract mean and standard deviation data for both the intervention and comparison teams. After screening titles and abstracts, 74 articles were included. Ensuring exhaustiveness in the study selection process and mitigating the risk of omitting significant, high-quality research, the full texts underwent an exhaustive evaluation during the second stage of screening. Notably, although robots can be considered a subclass of GAI, not all robots are capable of generating new content. Therefore, during the screening pro-

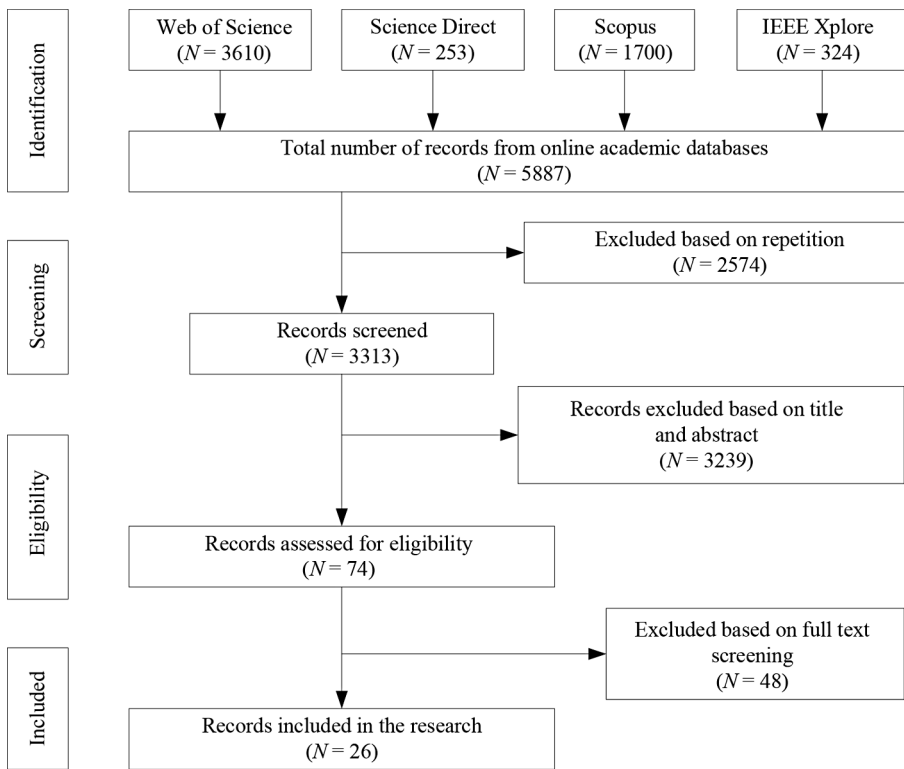


Fig. 1 A flowchart of literature inclusion

cess, experiments involving robots that lacked generative capabilities were excluded. The overall screening process involved independent judgment and selection by two researchers. The consistency between the two researchers in including literature was 0.93. The differences in literature inclusion were discussed and resolved by the two researchers. After completing the screening process, 26 articles were available for coding.

2.3 Coding procedures

To explore the impact of GAI on students' LO, we adopted the coding scheme proposed by Cooper (2016) and extracted the following information from 26 articles: (a) study title; (b) authors' names; (c) category of LO; (d) functional types of GAI; (e) educational levels; (f) intervention duration; (g) knowledge domains; (h) statistical results of LO (i.e., mean and standard deviation of the intervention and comparison teams); and (i) sample size.

Following the categorization schema of Salas-Pilco (2020), LO were categorized into PO, SO, and IO. In our previous examination of GAI functionalities, we found that GAI's utility in education primarily manifests in three aspects: assessing students' performance and abilities, providing tailored learning assistance, and serving as intelligent tutors in the classroom. Consequently, the functional types of GAI

were classified as personalized learning support, assessment, and intelligent tutoring systems in this study (Bahroun et al., 2023). For the educational levels, we adopted the classification strategy of Wu and Yu (2023), dividing them into primary education, secondary education, and higher education. Regarding the duration of a study, debates persist regarding the distinguishing criteria for categorizing interventions as extended versus brief, as noted by Ran et al. (2022). However, numerous studies recommend using 10 weeks as the threshold to distinguish between the two. Based on this, 10 weeks was adopted as the critical standard for coding in this study, with intervention duration either being less than 10 weeks (short-term intervention) or 10 weeks or more (long-term intervention). To further differentiate the knowledge domains involved in the experiments, we adopted UNESCO's (2018) International Standard Classification of Education, which encompasses various fields designated as "humanities and arts," "health and welfare," "agriculture," "social sciences, business and law," "services," "education," "engineering, manufacturing and construction," "general programs," "science," and "unspecified."

The coding was performed concurrently and autonomously by two researchers to guarantee the dependability and accuracy of the coding procedure. Any discrepancies that arose during the coding process were thoroughly discussed and resolved between the two researchers. The inter-coder reliability, measured at 0.91, signifies a strong degree of consistency between the two coders in their coding outcomes (Cohen, 1968). See Appendix A for coding details.

2.4 Statistical analysis and data synthesis

Meta-analysis is a method that gathers and combines numerical data from multiple studies on the same subject. After organizing the data, this study included a total of 26 articles, involving 44 comparisons, which means that there were 44 samples available for analysis. To further assess the impact of GAI on students' LO, the Comprehensive Meta-analysis software (version 3.0) (Biostat Inc., Englewood, NJ, USA) was utilized to analyze these experimental samples.

First, we looked at the funnel plot to visually check for publication bias. If bias was found, we used three tests – the Classic fail-safe N test, Begg's test, and Egger's test – to further assess this bias. Additionally, to determine whether differences between studies were caused by actual variance or by chance, a heterogeneity assessment was performed. The I^2 assists in interpreting this heterogeneity, with thresholds of 25% indicating low, 50% suggesting moderate, and 75% denoting high levels (Higgins et al., 2003). The random effects model is used when the studies come from different populations and there are real differences between them, rather than merely resulting from random variation. In contrast, the fixed effects model posits that the variability among studies originates from a common population, with discrepancies attributable to the sampling variability. In this study, the value of I^2 was 82.844%, which exceeds the threshold of 50%, so this analysis deemed the random effects model to be suitable (Higgins et al., 2011).

During the third phase, we determined the aggregate effect sizes (ES) and the mean ES of the encompassed studies. ES are important statistical measures (Fritz et al., 2012). ES in this study were used to quantify the actual impact of GAI on

students' LO, helping us to assess the effectiveness of GAI tools and make more targeted teaching recommendations and improvements. Since Hedges' g is more suitable for smaller samples, it was selected to measure ES in this study. According to Cohen's (1988) viewpoints, we considered ES near 0.20 to reflect a minor impact, those around 0.50 to indicate a moderate impact, and those exceeding 0.80 to signify a substantial impact. A significant Q -value ($Q=250.635$) indicates that the dispersion of ES is not just due to chance, so we needed to look into possible factors that might be influencing the results (Hedges & Olkin, 1985).

Concurrently, the presence of significant heterogeneity ($p=0.000$) compelled us to conduct a more in-depth analysis of the moderator variables to pinpoint what was causing this variation (Cortina, 2003). Additionally, we conducted subgroup analyses to explore the sources of these differences and to better understand the conditions under which GAI has varying effects on students' LO.

3 Results

3.1 Publication bias

To ensure the reliability of our findings, addressing publication bias is crucial, as it can skew the results if studies with certain outcomes are less likely to be published. In this meta-analysis, four methods were adopted to evaluate the potential for publication bias. We began by visually inspecting the funnel plot, which is a graphical representation that can indicate the presence of missing studies based on their results. The funnel plot appeared to be roughly symmetrical and the points were evenly distributed, suggesting a lower likelihood of publication bias within this study (see Fig. 2).

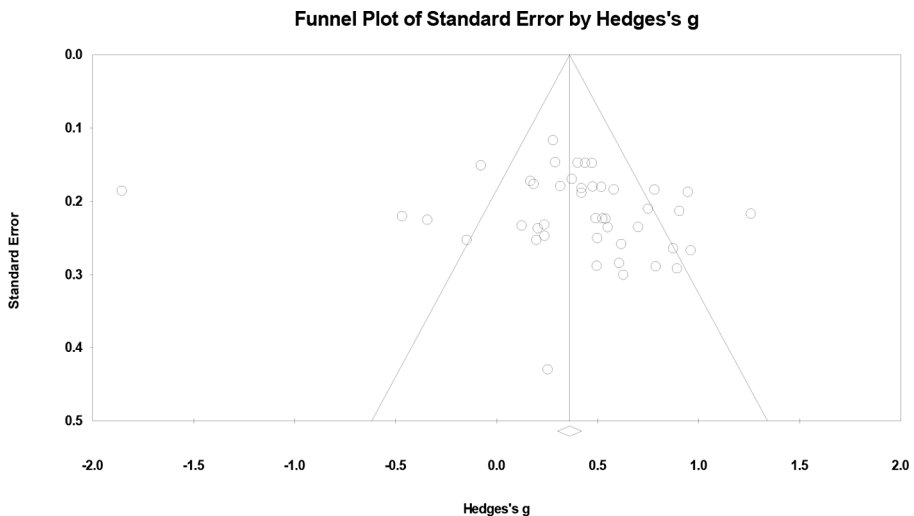


Fig. 2 Funnel plot of effect size data

Given that the outcomes of the funnel plot are susceptible to subjective interpretation, the assessment of publication bias should be corroborated by further analysis (Littell et al., 2008). The Classic fail-safe N test revealed that the results would remain robust even if there were at least 1,658 more unpublished articles, underscoring the strength of the findings. Similarly, Begg's test also suggested that the likelihood of publication bias in this study was statistically insignificant ($z=1.739$, $p>0.05$). Moreover, Egger's test, which specifically examines the asymmetry of the funnel plot, found the regression intercept to be close to zero and statistically insignificant (intercept=1.585, $p>0.05$), with a 95% confidence interval (CI) ranging from -1.482 to 4.653 further suggesting the lack of publication bias. The analysis yielded results that affirmed the robust internal validity of the ES and verified their suitability for encapsulating the scope of research conducted within the pertinent subject area.

3.2 Overall effects of GAI on students' LO

A central objective of this research endeavor was to grasp the intrinsic effects of GAI on students' LO. Figure 3 illustrates the forest plot that details the ES of the studies under review. The forest plot shows the range of likely values for each study's ES, known as the 95% CI, with square makers for individual study ES and horizontal

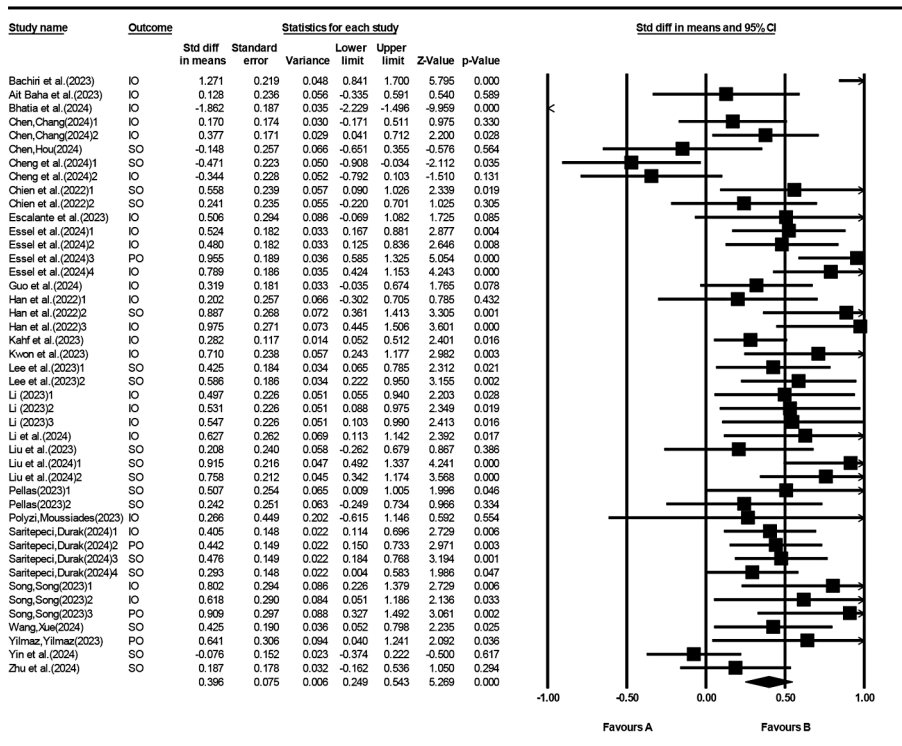


Fig. 3 Forest plot of the effect size for GAI on LO

bars for the CI. A diamond shape represents the overall ES when we combine all the studies.

Given the high level of heterogeneity ($I^2 = 82.844\%$), we employed a random effects model to calculate the overall ES across the examined research. The random effects model indicated an overall ES of 0.392 between GAI and students' LO (95%CI = [0.246, 0.538], $z = 5.264$, $p < 0.001$). Although this was a medium-to-small ES (Cohen, 1988), it confirmed that the intervention group exerted a beneficial influence on students' LO. This means that, compared to educational robots (Wu & Yu, 2023), the enduring influence of GAI on LO is nuanced and incremental. The heterogeneity in the detected effect across research was partially anticipated as a consequence of sampling variability.

The Q -test, which was used to check for consistency among the studies, was statistically significant ($Q = 250.635$, $p < 0.001$), indicating that most of the studies had similar ES (Borenstein et al., 2010). However, the I^2 value of 82.844% indicates that much of the variation we see is due to real differences in ES among the studies, not just random sampling (Borenstein et al., 2010). This high level of heterogeneity indicates that, in addition to sampling error and the influence of the actual ES of the studies, there may be one or more moderating factors responsible for the observed discrepancies in the data. Therefore, we proceeded with an analysis of moderating variables to investigate whether characteristics of the studies could explain differences in the estimated ES.

To evaluate the influence of GAI on LO of a diverse range of students, this study categorized LO into PO, SO, and IO. The ES calculations revealed that PO had the largest ES ($g = 0.701$, $N = 4$), reaching a medium level, while the ES for IO ($g = 0.372$, $N = 23$) and SO ($g = 0.347$, $N = 17$) were at a low level (see Table 1). It is noteworthy that GAI has a significant impact on all three categories of LO. Furthermore, we found that the heterogeneity for IO exceeded 75% ($I^2 = 88.443$), while the heterogeneity for PO and SO hovered around 50%. This indicates that for PO and SO, the performance of GAI in student learning exhibited higher homogeneity, which means that, like artificial intelligence robots, the function of GAI on students' PO and SO is less controversial or can be supported. The high heterogeneity observed in IO further underscores the importance and necessity of conducting an analysis of moderating variables.

3.3 Moderator analysis of the impact of GAI on students' LO

To understand the reasons behind the large differences we found among studies, we conducted an analysis of moderation effects. This type of analysis helped us find

Table 1 Overall effect size

	<i>N</i>	Hedges's <i>g</i>	Standard error	<i>z</i>	<i>p</i>	95%CI		I^2	Tau ²
						Lower	Upper		
IO	23	0.372	0.127	2.932	0.003	0.123	0.621	88.443	0.319
PO	4	0.701	0.144	4.868	0.000	0.419	0.983	42.584	0.035
SO	17	0.347	0.082	4.219	0.000	0.186	0.508	64.576	0.072

Note. *N*, the number of effect sizes

Table 2 Summary of the moderator analysis

Moderator variable level		<i>N</i>	Hedg- es's <i>g</i>	Stan- dard error	95%CI		<i>p</i>	Heterogeneity
					Lower	Upper		
Functional types of GAI	Assessment	6	0.429	0.172	0.091	0.766	0.013	$Q=0.110$
	Intelligent tutoring systems	7	0.277	0.432	-0.570	1.124	0.522	$p=0.946$
	Personalized learning support	31	0.418	0.058	0.303	0.532	0.000	
Educational levels	Higher education	30	0.394	0.096	0.206	0.583	0.000	$Q=15.247$
	Primary education	6	0.757	0.120	0.522	0.992	0.000	$p=0.000^{***}$
	Secondary education	8	0.112	0.114	-0.111	0.336	0.325	
Intervention duration	< 10 weeks	36	0.323	0.085	0.157	0.489	0.000	$Q=10.766$
	≥ 10 weeks	8	0.697	0.076	0.548	0.846	0.000	$p=0.001^{**}$
Knowledge domains	Education	17	0.484	0.069	0.349	0.620	0.000	$Q=11.394$
	Health and welfare	4	0.038	0.743	-1.418	1.494	0.959	$p=0.010^*$
	Humanities and arts	16	0.506	0.055	0.400	0.613	0.000	
	Science	7	0.046	0.129	-0.207	0.299	0.720	

Note. $*p<0.05$, $**p<0.01$, $***p<0.001$; *N*, the number of effect sizes; *Q*, heterogeneity statistic

factors that could explain why the studies varied so much. Given the considerable heterogeneity indicated by the *P*value ($P = 82.844\%$), we utilized a random effects model to investigate the influence of the potential moderating factors identified within this study, including functional types of GAI, educational levels, intervention duration, and knowledge domains. The latter three moderating variables showed significant effects. Table 2 presents the specific results for each moderating variable.

3.3.1 Functional types of GAI

As indicated in Table 2, no significant variation was observed in the impact of GAI on students' LO between the functional types of GAI ($p=0.946$). The *Q*-test ($Q(3)=0.110$) findings suggested that the variables did not exhibit significant heterogeneity with respect to the functional types of GAI. Specifically, the ES for assessment ($g=0.429$, $N=6$) and personalized learning support ($g=0.418$, $N=31$) were close to each other and approached a medium level of impact. In contrast, intelligent tutoring systems were noted to have the least impact on students' LO among the function types of GAI ($g=0.277$, $N=7$).

3.3.2 Educational levels

As depicted in Table 2, GAI was determined to exert a beneficial influence on student learning across different educational levels. Regarding the magnitude of the effect, GAI had the strongest impact on primary education students ($g=0.757$, $N=6$), followed by higher education ($g=0.394$, $N=30$), with secondary education experiencing the weakest effect ($g=0.112$, $N=8$). The statistical analysis revealed significant disparities among the various educational levels ($Q(3)=15.247$, $p<0.001$), indicating that the influence of GAI on students' LO varies with the level of education.

3.3.3 Intervention duration

In terms of intervention duration, 10 weeks is considered the threshold between short-term and long-term interventions. As can be seen from Table 2, GAI assistance in student learning for a duration of 10 weeks or more had a moderate impact ($g=0.697$, $N=8$), while the impact was smaller for interventions of less than 10 weeks ($g=0.323$, $N=36$). A significant discrepancy was identified with statistical significance, relating to the two sets of figures ($Q(2)=10.766$, $p<0.01$). This implies that the influence of GAI on students' LO varies depending on the duration of their GAI usage.

3.3.4 Knowledge domains

To evaluate the influence of GAI on students' LO across different knowledge domains, we categorized the studies based on the knowledge domains they covered into education, health and welfare, humanities and arts, and science. As depicted in Table 2, there are notable statistical disparities among the four factors ($Q(4)=11.394$, $p<0.05$). Specifically, humanities and arts achieved a medium ES ($g=0.506$, $N=16$), education was close to a medium ES ($g=0.484$, $N=17$), while the ES for health and welfare ($g=0.038$, $N=4$) and science ($g=0.046$, $N=7$) were extremely low.

3.4 Subgroup analyses

To better analyze how each factor affects different types of learning outcomes, we conducted subgroup analyses based on the following moderator variables: (1) functional types of GAI, (2) educational levels, (3) intervention duration, and (4) knowledge domains. It is worth noting that not all finer categories of subgroup divisions will have corresponding samples available. Therefore, this study reports on the data that had relevant sample information.

First, in terms of functional types of GAI, the impact on students' IO was higher for assessment ($g=0.469$, 95% CI = [0.156, 0.783]) and personalized learning support ($g=0.436$, 95% CI = [0.248, 0.624]) compared to intelligent tutoring systems ($g=0.176$, 95% CI = [-0.764, 1.116]). The intelligent tutoring systems function of GAI ($g=0.895$, 95% CI = [0.322, 1.468]) had the greatest impact on students' PO, but there was only one article in this category. As for SO, the impact on students' SO was quite similar for assessment ($g=0.375$, 95% CI = [-0.556, 1.306]) and personalized learning support ($g=0.354$, 95% CI = [0.195, 0.512]).

Second, regarding educational levels, GAI had the greatest impact on students' IO in primary education ($g=0.989$, 95% CI = [0.442, 1.536]), followed by higher education ($g=0.370$, 95% CI = [0.058, 0.682]), with secondary education having the smallest impact ($g=0.106$, 95% CI = [-0.178, 0.391]). Among the studies included, GAI showed a medium ES on the PO of students in higher education. Regarding the efficacy of GAI on students' SO, primary school students ($g=0.644$, 95% CI = [0.440, 0.848]) experienced the greatest benefits, whereas secondary school students ($g=0.124$, 95% CI = [-0.276, 0.524]) and those in higher education ($g=0.297$, 95% CI = [0.109, 0.485]) benefited relatively less.

In terms of intervention duration, the use of GAI for 10 weeks or more had a greater impact on students' IO ($g=0.626$, 95% CI = [0.456, 0.796]) and PO ($g=0.933$, 95% CI = [0.624, 1.243]). Regarding the impact of GAI on students' SO, all included studies had durations of GAI use of less than 10 weeks ($g=0.347$, 95% CI = [0.186, 0.508]).

Regarding knowledge domains, the highest ES for students' IO correspond to education ($g=0.613$, 95% CI = [0.428, 0.797]) and humanities and arts ($g=0.429$, 95% CI = [0.282, 0.576]), with eight studies in each field. For students' PO, GAI applications in humanities and arts ($g=0.895$, 95% CI = [0.322, 1.468]), education ($g=0.681$, 95% CI = [0.183, 1.180]), and science ($g=0.630$, 95% CI = [0.040, 1.220]) showed higher ES, but in these three cases, there is only one, two, and one article in each category, respectively. For students' SO, GAI applications in health and welfare ($g=0.876$, 95% CI = [0.356, 1.395]) and humanities and arts ($g=0.551$, 95% CI = [0.394, 0.709]) showed higher ES, with only one article in the health and welfare category.

4 Discussion

4.1 Effectiveness of GAI on students' LO

The research results suggested that students can achieve better LO with the long-term assistance of GAI. This result aligns with the findings of Wu and Yu (2023), Zhang et al. (2023), and Alemdag (2023). It further supports the view that GAI models such as artificial intelligence chatbots can improve LO (Mageira et al., 2022). The characteristics of GAI can explain why GAI exerts a beneficial influence on students' LO. Traditional artificial intelligence-driven tools and applications can help students with vocabulary memorization and classroom tests (Y. Wang et al., 2024), but these single-function carriers are becoming somewhat inadequate in diverse learning environments. The generative characteristics of GAI are sufficient to provide personalized guidance and assessment for students based on a variety of environments, starting from the students' needs to assist their multifaceted learning. Compared to social media tools that can enhance students' language abilities and cultural perspectives, GAI can provide immediate responses to students' behaviors and performances, offering real-time feedback (Gharaie, 2016), which can create an effective learning environment tailored to the students' learning states (Chen et al., 2010). In such an environment, students can interact and discuss more deeply with artificial intelligence chatbots, achieving the integration of new knowledge with existing knowledge (Lee et al., 2022), rather than waiting patiently for updates on relevant content videos or replies to others' comments.

Although this study revealed a small effect of GAI on students' LO, this impact is manifested through a combination of IO, PO, and SO. Therefore, while acknowledging the small but positive utility of long-term GAI intervention on students' LO, a more nuanced analysis within these specific categories is also required. The average ES across the three categories of LO were calculated in this study. Specifically, the integration of GAI had the greatest impact on students' PO ($g=0.701$). Traditional

chatbots or applications primarily respond to the questions or needs of questioners from an intellectual level, while GAI can offer students an abundance of training opportunities, enabling skill development through ongoing mimicry and practice (Hsu et al., 2021). For instance, to avoid the shyness and anxiety some students feel when speaking English in front of classmates and teachers, conversational chatbots offer a relaxed self-learning environment (Hsu et al., 2021; Jeon, 2022). Students can reduce the oral anxiety of language learners through constant dialogue and discussion with GAI (Han, 2020). This could be the reason why students make significant progress in PO.

To exercise students' motor skills and enhance their hands-on abilities, the intervention of GAI will be reasonable and effective. Although GAI showed a small effect on promoting students' IO and SO, its positive role was affirmative. Compared to traditional search engines such as Google, GAI is more popular among student groups, especially for seeking academic assistance, as it may provide them with unexpected new ideas and a more comprehensive framework for thinking (Zhang & Yang, 2024). The development of talent increasingly emphasizes the cultivation of creative thinking. Methods such as using Google are limited to "asking what" and "answering what," whereas GAI's targeted, general, and reflective responses to questions can stimulate students' self-efficacy, promote their thinking and innovation on the basis of the existing experience, and adjust their learning performance (Shahzad et al., 2024). Thus, in terms of IO and SO, GAI can provide students with additional ideas and support in the form of opinions.

4.2 Moderator variables of GAI on students' LO

The results indicated that there is no substantial statistical difference in how diverse GAI functionalities affect students' LO; instead, each of them aids in the enhancement of students' LO. Subgroup analysis results revealed that GAI's assessment and personalized learning support have a near-moderate positive impact on IO and SO. Compared to traditional artificial intelligence, GAI is poised to offer tailored support and aid to students by assessing a range of data, addressing their specific requirements (Luckin & Cukurova, 2019). For example, in interactions with students, GAI can express affirmation of students' viewpoints or supplement students' ideas more than search engines, thereby further enhancing students' self-efficacy and academic performance (Huang et al., 2022). The study has also demonstrated that the intelligent tutor systems feature of GAI has a large effect on PO. As a virtual online intelligent tutor, artificial intelligence chatbots enable students to select their favored approaches to learning (Yin et al., 2021) and provide explicit learning prompts to help students develop the good habit of completing their course assignments on time and accurately (Liu et al., 2022).

The intelligent tutor can also detect students' emotions and offer sufficient emotional communication, suggestions, and guidance according to students' emotions, thereby encouraging students to alleviate their own stress through specific behaviors (Klos et al., 2021). Although virtual reality technology may achieve this, the high cost of equipment and the high demands of the technology may make its use less satisfactory than that of GAI. Therefore, for students who want to practice practi-

cal skills or for courses aimed at promoting students' hands-on operational abilities, intelligent tutoring systems would be a good choice. Compared to human teachers, the biggest difference of GAI is its ability to systematically, repeatedly, and instantly provide detailed assessments and feedback on students' academic performance (such as writing scores), which greatly reduces students' cognitive load (Fidan & Gencel, 2022). Based on the above analysis, the positive impact of GAI with different functional focuses on students' LO seems understandable.

The impact of GAI varies across different educational levels. Although the integration of GAI has improved students' LO, its effect on middle school students is not significant. This supports Deveci Topal et al.'s (2021) viewpoint that GAI may not significantly enhance the academic performance of middle school students. Students in secondary education face more academic pressures from parents and schools compared to primary school and college students (Tang et al., 2020), and under such circumstances, they may be less able to focus on interacting with GAI for learning. Studies have determined that the influence of GAI is most pronounced in the LO of primary school students, particularly exerting a large effect on IO and a moderate effect on SO, which is inconsistent with the conclusions of Wu and Yu (2023). Jeon (2022) suggested that students in primary education may not interact effectively with chatbots and ChatGPT because their abilities in self-directed learning, language skills, and tool usage are weaker compared to those of middle and college students. However, researchers have found that younger students are more familiar with robots (Fernández-Llamas et al., 2018). Moreover, based on our observations, a growing body of empirical research starts with familiarizing students with the tools' operation prior to the experiment, aiding them in getting accustomed to utilizing robots or ChatGPT (e.g., Liang et al., 2024; Liu et al., 2024). Therefore, the notion that they cannot interact with GAI and do not know how to use it seems to be questionable. Studies have noted that many GAI designed for primary school students are presented in a game-based format (e.g., Bachiri et al., 2023; Liang et al., 2024), which greatly appeals to these students and further promotes their creative performance in SO. Primary school students generally maintain interest and curiosity in new things, which can further encourage them to improve their IO in the interaction without realizing it (Park et al., 2015).

It is evident that the impact of GAI on primary school students is not singular; its inherent functions, such as gamification and interactivity, can stimulate their interest while subtly enhancing their understanding of knowledge. Research has also found that GAI has a nearly moderate impact on college students' LO, especially in the area of PO. This may be because college students have stronger cognitive abilities and self-regulation skills, allowing them to maintain a higher level of engagement under task-driven conditions (Xu et al., 2022), thereby acquiring more skills during the task process. Thus, the long-term and continuous use of GAI will promote college students with a strong sense of autonomy to acquire more practical skills during the learning process.

Inconsistent with the research conclusions of Wu and Yu (2023), Deng and Yu (2023), and Wang et al. (2023), this research revealed that the duration of GAI implementation markedly affected learners' LO, with interventions lasting 10 weeks or more having a greater impact than those lasting less than 10 weeks. Previous evidence

suggested that the novelty effect of technology can briefly increase students' interest and engagement (Fryer et al., 2017). Nevertheless, a growing body of research has introduced GAI products to students before the intervention, familiarizing them with the technology to mitigate this novelty effect (e.g., Fidan & Gencel, 2022; Liu et al., 2024). Moreover, students in the information age can easily access different innovative technologies and are likely to become familiar with chatbots and ChatGPT in just a few minutes. Therefore, it is highly possible that by the time the formal experiment is conducted, the novelty effect for the students has already disappeared. Thus, what significantly correlates with students' LO is likely the learning experience and emotional support they obtain when GAI is integrated into their learning process. In the short term, GAI interventions are very likely to lead learners to experience negative emotions brought about by the technology, which will not be conducive to the enhancement of their LO (Pérez et al., 2020), or may show a slow promotion of their SO. In the long run, students may discover the positive effects of GAI on learning, gradually shifting from interest-driven learning to goal-driven learning, thereby further improving their LO. In this study, it has been demonstrated that GAI intervention periods of 10 weeks or longer have a moderate impact on students' IO and a large impact on PO. This further highlights the sustained influence of GAI intervention periods on the intellectual and physical growth of students. Long-term continuous use of GAI is beneficial for consolidating students' knowledge cognition and understanding, and frequent practice based on personalized learning scenarios can also deeply reinforce their skills.

Features of artificial intelligence such as ChatGPT may influence teaching and learning across different disciplines (Zhang & Tur, 2023). In this meta-analysis, the facilitative role of GAI in fields such as education, health and welfare, humanities and arts, and science is evident from the performance of students' LO. The impact of GAI on students' LO showed statistically significant differences across knowledge domains, contradicting the results reported by Wang et al. (2023). This could be attributed to the classification criteria we selected. Prior studies have shown that employing educational robots can boost students' academic performance in disciplines such as English and music (e.g., Kim, 2024; Zhou & Kim, 2024). However, there is also evidence that its role in computer science courses such as programming is quite limited (Kosar et al., 2024). It should be highlighted that, unlike ordinary chatbots, the efficacy of GAI hinges initially on its real-time adaptability (Mubin et al., 2013), which means that the selection and application of GAI may vary in specific disciplines. For instance, in oral training, the functions of GAI may be more inclined towards interaction, assessment, and feedback, while in mathematical problem scenarios, the functions of GAI will focus more on guidance and answering questions. To meet the learning objectives of integrating GAI into the classroom environment, the relevant robots, ChatGPT, or platforms must be utilized and analyzed within the scope of their principal functions. For instance, in the process of training nursing students, it is not enough to focus only on the impact of GAI on their grades; their practical abilities also need to be fully trained (Chang et al., 2021). Therefore, GAI in the form of intelligent tutoring systems may be appropriate and necessary for them. We also found that the application of GAI in the fields of education and humanities and arts had a moderate impact on students' IO, including not only the creation of

learning environments for specific subjects, but also guidance and feedback on student inquiries. Therefore, future research needs to design GAI products with different focuses according to different disciplines to meet the use of different GAI in a variety of learning activities.

5 Implications

5.1 Theoretical contributions and practical contributions

From a theoretical standpoint, this study has demonstrated a near moderate, statistically significant positive effect of GAI on students' LO. This means that the impact of GAI on student learning is indeed real, but more nuanced methods may be needed to study and assess its potential long-term effects. Although GAI plays a supportive role in improving student learning, it does not mean that GAI can completely replace teachers' teaching methods, which also provides some reference for properly positioning GAI. Comparing the impact of GAI on IO, PO, and SO, it has been demonstrated that GAI has a more pronounced effect on PO than on IO and SO. The four moderating variables of functional types of GAI, knowledge domains, intervention duration, and educational levels have been thoroughly discussed in this study, and it has been verified that the latter three factors are significant, providing a certain degree of parameter reference for the application of GAI.

Regarding practical application, this research provides valuable insights into how to use GAI to enhance students' LO and under what circumstances GAI should be employed to improve students' learning effectiveness. Specifically, in higher education courses involving fields such as education and humanities and arts, long-term (10 weeks and above) assistance from GAI can have a positive impact on students' IO and PO. For instance, advocating for college students to use ChatGPT over the long term, in a reasonable and regular manner, to complete classroom tasks and answer questions in quantitative research design classes can substantially foster the enhancement of their critical, reflective, and creative thinking skills (Essel et al., 2024).

Second, short-term (within 10 weeks) utilization of GAI for assisting learning has a good promotional effect on primary school learners' SO. Introducing artificial intelligence content generators (Lee et al., 2023) or chatbots (Liu et al., 2024) in primary school English classes and using them continuously in the short term is an interesting activity for students, as it can fully stimulate their interest in reading English books and gradually enhance their enjoyment of learning English. Although the performance in fostering learners' LO does not markedly vary among the three types of GAI functions, if the course focuses on cultivating students' motor skills, long-term assistance from intelligent tutor systems will more rapidly facilitate the mastery of skills.

This study not only provides a reference for the practical application of GAI, but also alerts us to the challenges that need to be considered in the specific application process of GAI. First, regardless of the educational level or subject for which GAI tools are used, the subscription and service fees they require are inevitable (Almasaad et al., 2024). At the same time, the research results advocate for college students

to use GAI for assistance and answering questions during task completion, but the adaptability and reliability of the feedback it provides still need further consideration. Since their responses come from a vast database, data-based responses may differ from students' personalized needs, thus failing to achieve the purpose of aiding learning.

Additionally, during the interaction process, there is also a risk that the conversation data between students and GAI tools could be incorporated into the data foundation, making data privacy something we need to weigh. For college students, although GAI has strong conversational and problem-solving capabilities, long-term use may indirectly increase students' dependency, reduce their motivation to learn, and thus lead to negative consequences of GAI (Ye et al., 2024). The results of this study indicate that for the primary school stage, teachers' understanding and moderate use of GAI tools are important and effective. Therefore, their acceptance of GAI and whether they have sufficient literacy to use it are important prerequisites for further applying GAI in the classroom.

5.2 Policy recommendations

This study confirms the positive effects of GAI on students' LO. With the support of GAI, significant changes will occur in the learning process and behaviors. When the application of GAI becomes too frequent or even mainstream, it may cause some disruption to education and teaching. This will prompt stakeholders to further develop policies and standards to ensure that the objectives of GAI use are aligned with our development direction.

This study, while providing a brief explanation of the application scenarios of GAI in teaching practice, needs to pay attention to the fact that the use of GAI should always be applied in education in a manner consistent with our values. To provide students with personalized content and feedback, the technical components of GAI applied in the classroom and their continuous updates, open-source, and effective models should be emphasized (Giannakos et al., 2024). At the same time, to prevent the leakage of students' personal data during interactions with GAI, the transparency and security of GAI tools need to be ensured. Educational institutions and the technical community also urgently need to establish new guidelines and governance frameworks for GAI to address algorithmic bias and data sensitivity issues, and to maximize the advantages of GAI in teaching practice with a responsible attitude (Mahrishi et al., 2024). The study proves GAI's usability but not its ability to replace teachers' and students' specific behaviors. Policymakers should view GAI as a complementary educational tool, rather than a substitute for interpersonal engagement. They must also construct ethical frameworks to direct the application of GAI, guaranteeing that it aligns with educational goals and upholds students' rights (Yusuf et al., 2024). Although the research results show that GAI encompasses diverse uses across the domains of science and humanities and arts, it is not considered a core subject in these fields (Chiu & Chai, 2020). Nowadays, the integration of artificial intelligence within school curricula has turned into a worldwide endeavor. Considering the potential long-term impact on and positive utility of GAI for students, future students and even teachers need to better understand GAI, make artificial intelligence

education a core subject in school education, and make GAI a true learning partner for both teachers and students (Chiu, 2023).

6 Conclusion

The ability of GAI to generate and automate processes significantly contributes to the student learning experience. This study synthesized findings from 26 empirical research studies to evaluate the influence of GAI on students' LO. The findings showed that the intervention of GAI exerts a beneficial impact on students' LO ($g=0.392$), although the impact is relatively small. This indicates that introducing GAI into students' learning to promote their thinking and growth is feasible, but it is also a lengthy process.

In the calculation of the average ES for the three types of LO, we found that GAI had a significant positive impact at a medium level on students' PO, while for IO and SO, there was a relatively smaller positive effect. Taking into account the significant variation among the studies included, this study used a moderator analysis to quantitatively assess four factors (i.e., functional types of GAI, knowledge domains, intervention duration, and educational levels). The first factor was found to be insignificant, while the latter three all showed significant statistical differences. It can be analyzed that using GAI in specific knowledge domains has varying degrees of impact on different types of students' LO. For example, compared to IO and SO, the application of GAI in the field of humanities and arts is more conducive to promoting students' PO. As the intervention time is extended, the impact of GAI on students' LO will gradually increase, particularly for IO and PO. GAI has a certain supportive role at different educational stages, especially during the primary school phase, where GAI assistance can effectively improve students' IO and SO.

This study conducted a comprehensive evaluation of the influence exerted by GAI on students' LO. However, there are still several inadequacies that need to be acknowledged. First, this study excluded some empirical research that did not provide sufficient statistical values, which may have influenced the findings from the meta-analysis (Cuijpers et al., 2017). Future researchers may further seek academic permission to access the data from the original authors of these empirical studies. Furthermore, the study was limited to four databases and English-language literature. Leveraging databases like Springer Link and Wiley Online Library, and broadening the search to encompass articles in additional languages such as Spanish and Korean would substantially augment the study pool for the meta-analysis.

An additional constraint of this meta-analysis is its narrow emphasis on a limited range of variables influencing the effectiveness of GAI interventions, and its neglect of other variables such as gender, cultural background, and learning preferences, which could also significantly affect students' LO. To enhance the positive effect of GAI on students, a comprehensive moderation analysis would yield a more holistic insight into the impact of GAI. Higher-order thinking skills are gradually becoming essential for students to face the future. Building on this study, future research that further narrows the scope of meta-analysis to the long-term impact of GAI on students' higher-order thinking skills will be able to provide effective strategies for

students to adapt to the development of the times. Additionally, this study focused on students' LO across all educational levels. The screening process revealed that the impact of GAI on students' LO is primarily evident in higher education settings. Therefore, a meta-analysis specifically tailored to a particular educational level would offer more appropriate significance and value for the use and impact of GAI.

7 Appendix A

Table a The studies selected for this meta-analysis

Author (year)	Outcome	Functional types of GAI	Educational levels	Intervention duration	Knowledge domains
Bachiri et al. (2023)	IO	Personalized learning support	Primary education	<10 weeks	Education
Ait Baha et al. (2024)	IO	Personalized learning support	Secondary education	<10 weeks	Science
Bhatia et al. (2024)	IO	Intelligent tutoring systems	Higher education	<10 weeks	Health and welfare
Chen and Chang (2024) [1]	IO	Personalized learning support	Secondary education	<10 weeks	Science
Chen and Chang (2024) [2]	IO	Personalized learning support	Secondary education	<10 weeks	Science
Chen and Hou (2024)	SO	Personalized learning support	Higher education	<10 weeks	Education
Cheng et al. (2024) [1]	SO	Personalized learning support	Secondary education	<10 weeks	Science
Cheng et al. (2024) [2]	IO	Personalized learning support	Secondary education	<10 weeks	Science
Chien et al. (2022) [1]	SO	Personalized learning support	Secondary education	<10 weeks	Humanities and arts
Chien et al. (2022) [2]	SO	Personalized learning support	Secondary education	<10 weeks	Humanities and arts
Escalante et al. (2023)	IO	Assessment	Higher education	<10 weeks	Humanities and arts
Essel et al. (2024) [1]	IO	Personalized learning support	Higher education	≥10 weeks	Education
Essel et al. (2024) [2]	IO	Personalized learning support	Higher education	≥10 weeks	Education
Essel et al. (2024) [3]	PO	Personalized learning support	Higher education	≥10 weeks	Education
Essel et al. (2024) [4]	IO	Personalized learning support	Higher education	≥10 weeks	Education
Guo et al. (2024)	IO	Assessment	Higher education	<10 weeks	Humanities and arts
Han et al. (2022) [1]	IO	Assessment	Higher education	<10 weeks	Health and welfare
Han et al. (2022) [2]	SO	Assessment	Higher education	<10 weeks	Health and welfare
Han et al. (2022) [3]	IO	Assessment	Higher education	<10 weeks	Health and welfare

Table a The studies selected for this meta-analysis

Author (year)	Outcome	Functional types of GAI	Educational levels	Intervention duration	Knowledge domains
Kahf et al. (2023)	IO	Personalized learning support	Higher education	<10 weeks	Humanities and arts
Kwon et al. (2023)	IO	Personalized learning support	Primary education	≥10 weeks	Humanities and arts
Lee et al. (2023) [1]	SO	Personalized learning support	Primary education	<10 weeks	Humanities and arts
Lee et al. (2023) [2]	SO	Personalized learning support	Primary education	<10 weeks	Humanities and arts
Li (2023) [1]	IO	Intelligent tutoring systems	Higher education	<10 weeks	Education
Li (2023) [2]	IO	Intelligent tutoring systems	Higher education	<10 weeks	Education
Li (2023) [3]	IO	Intelligent tutoring systems	Higher education	<10 weeks	Education
Li et al. (2024)	IO	Personalized learning support	Higher education	<10 weeks	Humanities and arts
Liu et al. (2023)	SO	Personalized learning support	Higher education	<10 weeks	Education
Liu et al. (2024) [1]	SO	Personalized learning support	Primary education	<10 weeks	Humanities and arts
Liu et al. (2024) [2]	SO	Personalized learning support	Primary education	<10 weeks	Humanities and arts
Pellas (2023) [1]	SO	Personalized learning support	Higher education	<10 weeks	Education
Pellas (2023) [2]	SO	Personalized learning support	Higher education	<10 weeks	Education
Polyzi and Moussiades (2023)	IO	Personalized learning support	Higher education	<10 weeks	Humanities and arts
Saritepeci and Durak (2024) [1]	IO	Personalized learning support	Higher education	<10 weeks	Education
Saritepeci and Durak (2024) [2]	PO	Personalized learning support	Higher education	<10 weeks	Education
Saritepeci and Durak (2024) [3]	SO	Personalized learning support	Higher education	<10 weeks	Education
Saritepeci and Durak (2024) [4]	SO	Personalized learning support	Higher education	<10 weeks	Education
Song and Song (2023) [1]	IO	Intelligent tutoring systems	Higher education	≥10 weeks	Humanities and arts
Song and Song (2023) [2]	IO	Intelligent tutoring systems	Higher education	≥10 weeks	Humanities and arts
Song and Song (2023) [3]	PO	Intelligent tutoring systems	Higher education	≥10 weeks	Humanities and arts
Wang and Xue (2024)	SO	Personalized learning support	Higher education	<10 weeks	Humanities and arts
Yilmaz and Yilmaz (2023)	PO	Personalized learning support	Higher education	<10 weeks	Science
Yin et al. (2024)	SO	Assessment	Higher education	<10 weeks	Science
Zhu et al. (2024)	SO	Personalized learning support	Secondary education	<10 weeks	Education

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Declarations

Conflict of interest Not applicable.

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Authors and Affiliations

Yinkun Zhu¹ · Qiwen Liu¹ · Li Zhao¹ 

✉ Li Zhao
li.zhao@njnu.edu.cn

Yinkun Zhu
230602157@njnu.edu.cn

Qiwen Liu
240602178@njnu.edu.cn

¹ Adolescent Education and Intelligence Support Lab of Nanjing Normal University, Laboratory of Philosophy and Social Sciences at Universities in Jiangsu Province, 122 Ninghai Road, Nanjing, Jiangsu 210097, China